

Lecture	T/Th 3:30-5:30, UW1-050
Instructor	Prof Pisan Email: pisan@uw.edu
Office	UW1-260Q Phone: 425.352.3741
Office Hours	W 2-3pm, Th 2-3pm or by appointment

Course Description

This fast-paced course is intended to enable students to design solutions to programming problems using object-oriented techniques. The course integrates the fundamental discrete mathematics aspects of computing with detailed instruction in end-to-end software design. By the end of this quarter, you will be familiar with much of the C++ language and the basics of object-oriented programming. You will understand how to analyze a problem and design a solution. You will understand the basic data structures and algorithms which are the basis for much of computer science. You will also be able to analyze the trade-offs among memory, execution time, and implementation complexity associated with them. Topics include: recursion, computational complexity and algorithm analysis, logic, induction, lists, stacks, queues, sorting and searching. Also covered are object oriented fundamentals such as abstraction, encapsulation, hierarchy, and polymorphism.

Textbooks

Data Abstraction and Problem Solving with C++: Walls & Mirrors (7th Edition), Frank M. Carrano,
Addison-Wesley.

An Active Introduction to Discrete Mathematics and Algorithms (version 2.6), Charles Cusack, David Santos, GNU Free Software, 2016. ([available online](https://github.com/pisanorg/w/wiki))

For additional books and references see <https://github.com/pisanorg/w/wiki>

Grading

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. 2.0 roughly corresponds to 75%

Quizzes and in-class participation: 10%

Assignments: 35%

Midterm: 25%

Comprehensive Final Exam: 30%

Policies

Attendance: Attend all classes. If you are going to miss a class, let me know via email. If you miss a class, contact your classmates to find out what you missed. If you miss more than one class, come to my office hours to explain your circumstances.

Electronics: Minimal electronics in class unless we have a specific class exercise that requires a computer. Research shows that using pen and paper to take notes is much more effective than using a computer. Computers are often distracting to you and also to others around you. If you need accommodations for taking notes, let me know. If you need to send a text message or take a phone call, step outside to do it.

Assignment Submission: Strict deadlines, but everybody occasionally slips, so you can submit 1 assignment up to 24 hours late for any reason (email me to let me know). All other assignments have to be submitted on time. If you need an extension due to exceptional circumstances, email or talk to me before the due date. Exceptional circumstances only; work, other classes, sleep, traffic, etc are not exceptional.

Effort: Expect to spend 10-15 hours per week outside class. If an assignment is worth 10%, expect to spend 10-20 hours on the assignment. If you are spending too much time or too little time, let me know we'll adjust the course content. Learning happens best when you are challenged and get to stretch your limits.

Academic Integrity: Do the right thing, see <http://guides.lib.uw.edu/bothell/ai> for details on what is "right" if in doubt. Talking about code is OK, looking at each others code is not OK. Looking at references to understand how a function gets used is OK; looking up assignment solutions is not OK.

Communication: Join the course channel on discord <https://discord.gg/5mEm92e> . Discord is an extension of the classroom, so use a meaningful nickname and act professionally. If your question can be answered by a classmate, post it to discord rather than using email or waiting for office hours. Office hours are for complex issues or topics you are struggling with.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Access and Accommodations: Your experience in this class is important to me. If you have already established accommodations with Disability Resources for Students (DRS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course. See <http://www.uwb.edu/studentaffairs/drs> if you need to establish accommodations.

Religious Accommodations: Washington state law requires that UW develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including more information about how to request an accommodation, is available at Faculty Syllabus Guidelines and Resources. Accommodations must be requested within the first two weeks of this course using the Religious Accommodations Request form available at <https://registrar.washington.edu/students/religious-accommodations-request/>

Common Course Policies for the School of STEM: See <https://www.uwb.edu/stem/about/stem-policies> for additional information on Academic Integrity, Access and Accommodations, Classroom Emergency Preparedness, For Our Veterans, Grade of Incomplete, Inclement Weather, Parenting Resources, Respect for Diversity. Student Support Services, Wondering How to Address Faculty? etc. (P.S. I

prefer to be addressed as 'Professor Pisan')

Course Material: Lecture notes and other material will be posted to Canvas under "Files" the week after the lecture.

Topics

Week 1:

- Ch 1: Data Abstraction: The Walls
- Appendix A: Review of C++ Fundamentals
- Appendix K: C++ for Java Programmers

Week 2:

- Ch 3: Array-Based Implementations
- Interlude 1.1: A Problem to Solve
- Interlude 1.2: Implementing a Solution
- Interlude 6: Overloaded Operators and Friend Access

Week 3:

- Ch 4: Link-Based Implementations
- Interlude 1.3: Templates

Week 4:

- Ch 2: Recursion- The Mirrors
- Ch 5: Recursion as a Problem Solving Technique
- Appendix E: Mathematical Inductions

Week 5: No Class Thursday Oct 24th, exercises/video will be assigned

- Ch 6: Stacks
- Ch 7: Implementation of the ADT Stack
- Ch 8: Lists

Week 6:

- Ch 9: List Implementations
- Ch 10: Algorithm Efficiency

Week 7:

- Ch 10: Algorithm Efficiency
- Ch 11: Sorting Algorithms

Week 8:

- Cusak Chapter 7: Algorithm Analysis

Week 9:

- Ch 13: Queues and Priority Queues
- Ch 14: Queue Implementations

Week 10: No Class Thursday Nov 28th, Thanksgiving

- Ch 15: Trees

Week 11: Review

Week 12: Final exam: Tuesday, Dec 10th, 3:30pm